Ericsson PE Routing Protocol Flap Failure SOP for Node ID: JAY-PE-09 at Jayanagar Metro Site

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_1573

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, and ISIS LSP flooding) failure on a Ericsson PE (Provider Edge) node, Node ID: JAY-PE-09, located at Jayanagar Metro Site. The failure is characterized by the alarm code ALARM_1573 with a severity level of Critical. The underlying cause of this failure domain is the complex interaction between the routing protocols (BGP, OSPF, and ISIS) and the hardware components on the PE node. The protocols continuously exchange routing information and maintain routing tables to ensure optimal routing and connectivity. However, during periods of high network activity or unexpected changes in the network topology, the protocols may become unstable, leading to route flapping. This can cause network instability, increased CPU utilization, and potential network outages.

Pre-requisite Checks

- Verify that the redundant link is up and active before proceeding.

- Ensure that remote hands are dispatched to the Jayanagar Metro Site.

- Check the Ericsson node's CPU and memory utilization to determine if there are any resource constraints.

Potential Root Causes

Cause	Probability
BGP route churn caused by misconfigured AS-PATH or Communities	High
OSPF adjacency loss due to network partitioning or interface flap	Medium
ISIS LSP flooding caused by misconfigured metrics or priority	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary | include Route-Flap, Expected Output: Route-Flap: 10, Flap-Time: 30s, Description: This command isolates the BGP route flapping issue by displaying the number of route flaps and the time since the last flap.

Command: show ospf interface | include adjacency, Expected Output: Adjacency: DOWN, Description: This command identifies the OSPF adjacency loss issue by displaying the interface status and adjacency state.

Command: show isis database | include LSP, Expected Output: LSP: 10, flooded: 5, Description: This command isolates the ISIS LSP flooding
```

```
issue by displaying the number of LSPs and the number of flooded LSPs.
```

Resolution Steps

Step 1: Step: Verify and correct BGP AS-PATH and Communities configuration, Command: verify bgp as-path communities, Description: Verify the BGP AS-PATH and Communities configuration to ensure it is accurate and consistent across all AS-PATHs and Communities.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf adjacency, Description: Reset the OSPF adjacency to re-establish the connection and resolve the adjacency loss issue.

Step 3: Step: Adjust ISIS metrics and priority, Command: adjust isis metrics priority, Description: Adjust the ISIS metrics and priority to ensure optimal routing and prevent LSP flooding.

Post-Check Verification

```
Command: show bgp summary | include Route-Flap, Expected Output: Route-Flap: 0, Flap-Time: 0s, Description: Verify that the BGP route flapping issue has been resolved by displaying the number of route flaps and the time since the last flap.

Command: show ospf interface | include adjacency, Expected Output: Adjacency: UP, Description: Verify that the OSPF adjacency loss issue has been resolved by displaying the interface status and adjacency state.

Command: show isis database | include LSP, Expected Output: LSP: 0, flooded: 0, Description: Verify that the ISIS LSP flooding issue has been resolved by displaying the number of LSPs and the number of flooded LSPs.
```

Roll-back Procedures

- Step: Restore original BGP AS-PATH and Communities configuration, Command: restore bgp as-path communities, Description: Restore the original BGP AS-PATH and Communities configuration to ensure consistency and accuracy.

- Step: Reset OSPF adjacency to original state, Command: reset ospf adjacency, Description: Reset the OSPF adjacency to its original state to resolve any issues caused by the resolution steps.

- Step: Revert ISIS metrics and priority to original values, Command: revert isis metrics priority, Description: Revert the ISIS metrics and priority to their original values to ensure optimal routing and prevent LSP flooding.